

7. Trazas de Wireshark

Captura del intercambio de DNS Query y DNS Reply de la petición desde el cliente a nuestro DNS primario mediante dig `www.nombreorganizacion6535.org`:

1	0.000000000	192.168.230.4	192.168.230.5	DNS	101 Standard query 0x882b A www.nombreorganizacion6535.org OPT
2	0.002838703	192.168.230.5	192.168.230.4	DNS	196 Standard query response 0x882b A www.nombreorganizacion6535.org A 192.168.230.5 NS servidor.nombreorganizacion6535...

imagen 49: intercambio DNS Query y DNS Reply

También aquí tenemos la captura del protocolo SMTP mediante el envío de correos usando Thunderbird:

14	19.061631308	192.168.230.4	192.168.230.5	SMTP	88 C: EHLO [192.168.230.4]
15	19.063819748	192.168.230.5	192.168.230.4	TCP	66 25 → 50768 [ACK] Seq=71 Ack=23 Win=29056 Len=0 TSval=2587837 TSecr=2528339
16	19.064666349	192.168.230.5	192.168.230.4	SMTP	186 S: 250-server1 Hello [192.168.230.4] [192.168.230.4] SIZE 52428800 8BITIME PIPELINING PRDR HELP
17	19.064682363	192.168.230.4	192.168.230.5	TCP	66 50768 → 25 [ACK] Seq=23 Ack=191 Win=29312 Len=0 TSval=2528340 TSecr=2587837
18	19.125156431	192.168.230.4	192.168.230.5	SMTP	138 C: MAIL FROM:<usuario1@nombreorganizacion6535.org> BODY=8BITIME SIZE=481
19	19.128133548	192.168.230.5	192.168.230.4	SMTP	74 S: 250 OK
20	19.128163698	192.168.230.4	192.168.230.5	TCP	66 50768 → 25 [ACK] Seq=95 Ack=199 Win=29312 Len=0 TSval=2528356 TSecr=2587853
21	19.557206055	192.168.230.4	192.168.230.5	SMTP	113 C: RCPT TO:<usuario2@nombreorganizacion6535.org>
22	19.560191624	192.168.230.5	192.168.230.4	SMTP	80 S: 250 Accepted
23	19.560222585	192.168.230.4	192.168.230.5	TCP	66 50768 → 25 [ACK] Seq=142 Ack=213 Win=29312 Len=0 TSval=2528464 TSecr=2587961
24	19.567147029	192.168.230.4	192.168.230.5	SMTP	72 C: DATA
25	19.568908864	192.168.230.5	192.168.230.4	SMTP	122 S: 354 Enter message, ending with "." on a line by itself

25	19.568908864	192.168.230.5	192.168.230.4	SMTP	122 S: 354 Enter message, ending with "." on a line by itself
26	19.586741101	192.168.230.4	192.168.230.5	SMTP	547 C: DATA fragment, 481 bytes
27	19.588003165	192.168.230.4	192.168.230.5	SMTP/I...	69 from: usuario1 <usuario1@nombreorganizacion6535.org>, subject: Prueba V2, (text/plain)
28	19.591818240	192.168.230.5	192.168.230.4	TCP	66 25 → 50768 [ACK] Seq=269 Ack=632 Win=30080 Len=0 TSval=2587968 TSecr=2528470
29	19.623545966	192.168.230.5	192.168.230.4	SMTP	94 S: 250 OK id=1w0lzk-00012v-28
30	19.627877464	192.168.230.4	192.168.230.5	SMTP	72 C: QUIT

imágenes 50 y 51: trazas del protocolo SMTP

Captura de dig `MX nombreorganizacion6535.org`:

No.	Time	Source	Destination	Protocol	Length	Info
9	62.149670393	192.168.56.101	192.168.56.102	DNS	97	Standard query 0x6a6c MX nombreorganizacion6535.org OPT
10	62.150791195	192.168.56.102	192.168.56.101	DNS	219	Standard query response 0x6a6c MX nombreorganizacion6535.org MX 10 smtp.nombreorganizacion6535.org NS servidor.nombreorganizacion6535...

imagen 52: captura de dig MX nombreorganizacion6535.org

Se puede ver como el cliente contacta con el servidor y este le responde de manera simple.

Captura de dig `www.um.es`:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.56.101	192.168.56.102	DNS	82	Standard query 0xdc42 A www.um.es OPT
2	0.000582	10.0.2.15	192.203.230.10	DNS	82	Standard query 0xa495 A www.um.es OPT
4	0.010501	192.203.230.10	10.0.2.15	DNS	549	Standard query response 0xa495 A www.um.es NS a.nic.es NS c.nic.es NS g.nic.es NS h.nic.es DS RRSIG A 194.69.254.1 A 194.0.34.53 A 20...
6	0.010750	10.0.2.15	194.0.33.53	DNS	82	Standard query 0xde8d A www.um.es OPT
18	0.076466	194.0.33.53	10.0.2.15	DNS	82	Standard query response 0xde8d A www.um.es OPT
22	0.142783	10.0.2.15	194.0.33.53	DNS	96	Standard query 0x043d A www.um.es OPT
24	0.208834	194.0.33.53	10.0.2.15	DNS	1115	Standard query response 0x043d A www.um.es NS sun.rediris.es NS chico.rediris.es NS dns2.um.es NS dns1.um.es NSEC3 RRSIG NSEC3 RRSIG ...
26	0.210009	fd17:625c:f037:2:a0...	2001:720:1710:1::2	DNS	102	Standard query 0xb995 A www.um.es OPT
28	0.210873	fe80::2	fd17:625c:f037:2:a0...	ICMPv6	150	Destination Unreachable (No route to destination)
33	1.010150	10.0.2.15	155.54.1.2	DNS	82	Standard query 0x1236 A www.um.es OPT
34	1.037411	155.54.1.2	10.0.2.15	DNS	119	Standard query response 0x1236 A www.um.es CNAME wwwclu.um.es A 155.54.212.103 OPT
37	1.081942	192.168.56.102	192.168.56.101	DNS	379	Standard query response 0xdc42 A www.um.es CNAME wwwclu.um.es A 155.54.212.103 NS sun.rediris.es NS chico.rediris.es NS dns2.um.es NS...

imagen 53: captura de dig www.um.es

En esta captura se puede observar como va pidiendo de manera iterativa los recursos que necesita empezando por `.es`, después `um.es` y finalmente `www.um.es`.

Captura de recibo de correo:

No.	Time	Source	Destination	Protocol	Length	Info
8	0.012031728	192.168.56.102	192.168.56.101	POP	86	S: +OK Dovecot ready.
10	0.012393094	192.168.56.101	192.168.56.102	POP	72	C: CAPA
12	0.013614513	192.168.56.102	192.168.56.101	POP	149	S: +OK
13	0.014006699	192.168.56.101	192.168.56.102	POP	78	C: AUTH PLAIN
14	0.015755647	192.168.56.102	192.168.56.101	POP/IMF	70	+
15	0.015955972	192.168.56.101	192.168.56.102	POP	92	C: AHVzdWfYaw8yAHVzdWfYaw8y
16	0.030511584	192.168.56.102	192.168.56.101	POP	82	S: +OK Logged in.
17	0.030713877	192.168.56.101	192.168.56.102	POP	72	C: STAT
18	0.040693979	192.168.56.102	192.168.56.101	POP	77	S: +OK 1 844
19	0.042230334	192.168.56.101	192.168.56.102	POP	72	C: LIST
20	0.044053202	192.168.56.102	192.168.56.101	POP	93	S: +OK 1 messages:
21	0.045081078	192.168.56.101	192.168.56.102	POP	72	C: UIDL
22	0.046384958	192.168.56.102	192.168.56.101	POP	94	S: +OK
23	0.046898692	192.168.56.101	192.168.56.102	POP	74	C: RETR 1
24	0.048182359	192.168.56.102	192.168.56.101	POP	929	S: +OK 844 octets
25	0.069944262	192.168.56.101	192.168.56.102	POP	72	C: QUIT
26	0.076525947	192.168.56.102	192.168.56.101	POP	84	S: +OK Logging out.

imagen 54: captura de recibo de correo

Aquí se ve como se pide el recurso y se envía correctamente.

Trazas del intercambio de mensajes tras haber implementado IPSec y hacer un ping a nuestro servidor (imagen 51) y al acceder a la página web con el servicio Apache activo (imagen 52).

No.	Time	Source	Destination	Protocol	Length	Info
2	7.425495208	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request ...
3	7.426405354	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
4	8.427316489	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
5	8.428486315	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
6	9.429883885	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
7	9.430737513	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
8	10.436583494	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
9	10.438022648	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
10	11.443268538	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
11	11.444945065	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
14	12.446981025	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
15	12.448535512	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
16	13.448453010	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
17	13.450652896	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
18	14.450517682	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
19	14.451307009	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
20	15.452330233	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
21	15.453255623	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
22	16.458942051	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
23	16.462837405	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply
24	17.461070270	192.168.230.4	192.168.230.5	ICMP	122	Echo (ping) request
25	17.462293374	192.168.230.5	192.168.230.4	ICMP	122	Echo (ping) reply

Reserved: 0000
AH SPI: 0xc9e00be9
AH Sequence: 35
AH ICV: ddd074c7ce2a01918e1e143e

0000	08 00 27	da a9 89 08 00	27 88 18 17 08 00	45 00	E.
0010	00 6c 93	f0 40 00 40 33	59 13 c0 a8 e6 04	c0 a8	..@.3 Y.....

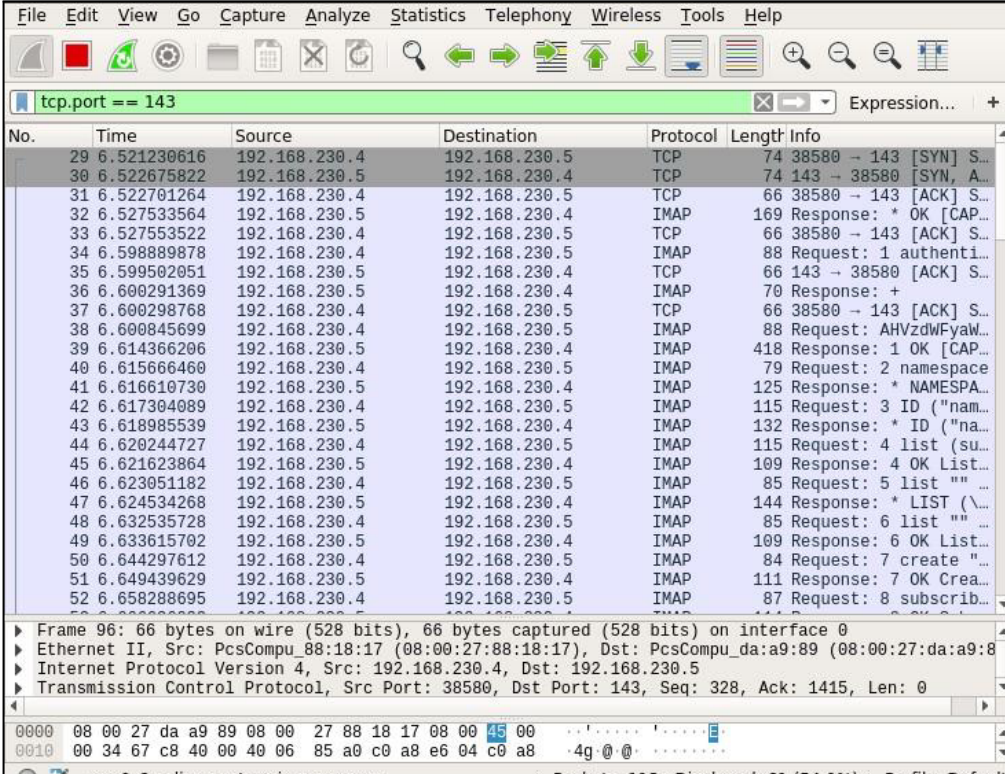
Ready to load or capture Packets: 25 · Displayed: 22 (88.0%) Profile: Default

imagen 55: trazas de verificación del funcionamiento de IPSec con AH

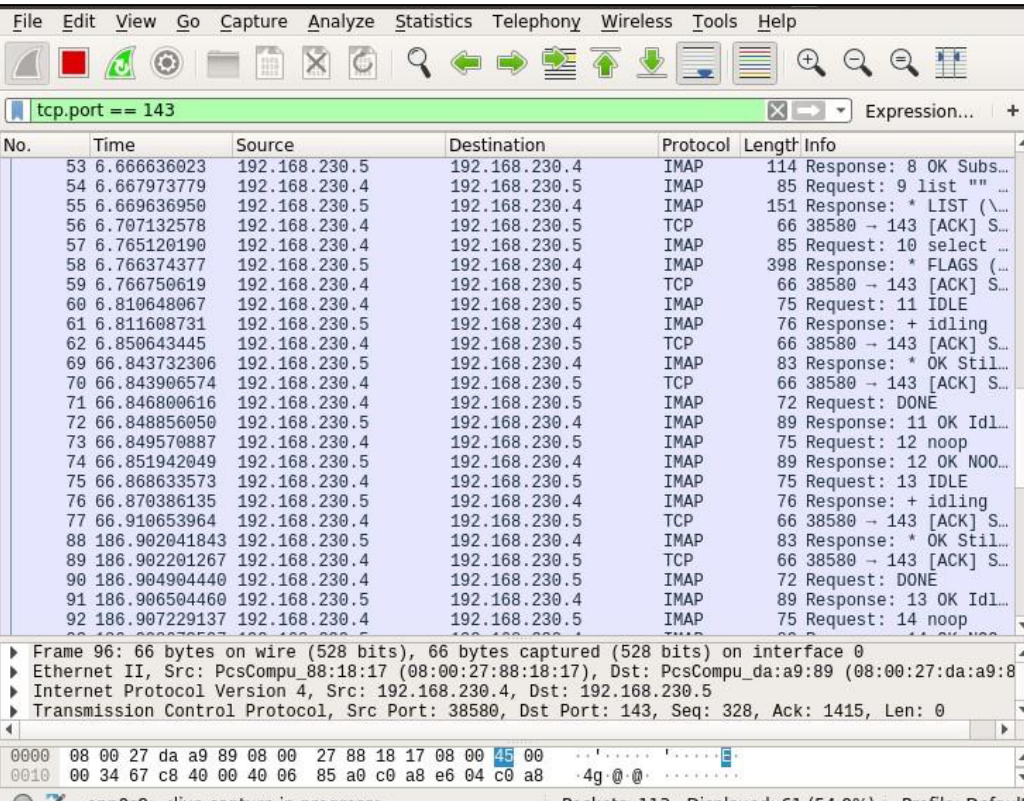
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
isakmp ah						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.230.4	192.168.230.5	DNS	114	Standard query...
2	0.000628414	192.168.230.4	192.168.230.5	DNS	114	Standard query...
3	0.001826740	192.168.230.5	192.168.230.4	DNS	209	Standard query...
4	0.001851588	192.168.230.5	192.168.230.4	DNS	165	Standard query...
5	0.012125860	192.168.230.4	192.168.230.5	TCP	98	34450 → 443 [S...
6	0.012948150	192.168.230.5	192.168.230.4	TCP	98	443 → 34450 [S...
7	0.012975687	192.168.230.4	192.168.230.5	TCP	90	34450 → 443 [A...
8	0.013736787	192.168.230.4	192.168.230.5	TLSv1.2	299	Client Hello
9	0.014925527	192.168.230.5	192.168.230.4	TCP	90	443 → 34450 [A...
10	0.019165952	192.168.230.5	192.168.230.4	TLSv1.2	1514	Server Hello
11	0.019200627	192.168.230.4	192.168.230.5	TCP	90	34450 → 443 [A...
12	0.019311454	192.168.230.5	192.168.230.4	TLSv1.2	1193	Certificate, S...
13	0.019327995	192.168.230.4	192.168.230.5	TCP	90	34450 → 443 [A...
14	1.896775496	192.168.230.4	192.168.230.5	DNS	114	Standard query...
15	1.899016810	192.168.230.5	192.168.230.4	DNS	499	Standard query...
16	1.901909554	192.168.230.4	192.168.230.5	DNS	114	Standard query...
17	1.903855929	192.168.230.5	192.168.230.4	DNS	252	Standard query...
18	1.917046267	192.168.230.4	192.168.230.5	DNS	114	Standard query...
19	1.918221518	192.168.230.4	192.168.230.5	DNS	114	Standard query...
20	1.919756792	192.168.230.5	192.168.230.4	DNS	499	Standard query...
21	1.920734493	192.168.230.5	192.168.230.4	DNS	252	Standard query...
22	1.930300406	192.168.230.4	192.168.230.5	DNS	114	Standard query...
23	1.932561127	192.168.230.5	192.168.230.4	DNS	499	Standard query...
24	1.937124400	192.168.230.4	192.168.230.5	DNS	114	Standard query...
25	1.938337350	192.168.230.5	192.168.230.4	DNS	252	Standard query...
Reserved: 0000						
AH SPI: 0xc665d962						
AH Sequence: 196						
AH ICV: 7c67cb0b5d05aa80a5d242d1						
0000 08 00 27 da a9 89 08 00 27 88 18 17 08 00 45 00 ..'. '.....E.						
0010 01 1d ec ed 40 00 40 33 ff 64 c0 a8 e6 04 c0 a8 ...@.03 .d.....						
wireshark enp...Mla95C.pcapn Packets: 104 · Displayed: 104 (100.0%) · Dropped: 0 (0.0%) Profile: Default						

imagen 56: visualización de todos los protocolos funcionando conjuntamente

Captura de los intercambios entre el servidor de correo IMAP y el cliente:



No.	Time	Source	Destination	Protocol	Length	Info
29	6.521230616	192.168.230.4	192.168.230.5	TCP	74	38580 → 143 [SYN] S...
30	6.522675822	192.168.230.5	192.168.230.4	TCP	74	143 → 38580 [SYN, A...
31	6.522701264	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
32	6.527533564	192.168.230.5	192.168.230.4	IMAP	169	Response: * OK [CAP...
33	6.527553522	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
34	6.598889878	192.168.230.4	192.168.230.5	IMAP	88	Request: 1 authenti...
35	6.599502051	192.168.230.5	192.168.230.4	TCP	66	143 → 38580 [ACK] S...
36	6.600291369	192.168.230.5	192.168.230.4	IMAP	70	Response: +
37	6.600298768	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
38	6.600845699	192.168.230.4	192.168.230.5	IMAP	88	Request: AHVzdWYyaW...
39	6.614366206	192.168.230.5	192.168.230.4	IMAP	418	Response: 1 OK [CAP...
40	6.615666460	192.168.230.4	192.168.230.5	IMAP	79	Request: 2 namespace
41	6.616610730	192.168.230.5	192.168.230.4	IMAP	125	Response: * NAMESPA...
42	6.617304089	192.168.230.4	192.168.230.5	IMAP	115	Request: 3 ID ("nam...
43	6.618985539	192.168.230.5	192.168.230.4	IMAP	132	Response: * ID ("na...
44	6.620244727	192.168.230.4	192.168.230.5	IMAP	115	Request: 4 list (su...
45	6.621623864	192.168.230.5	192.168.230.4	IMAP	109	Response: 4 OK List...
46	6.623051182	192.168.230.4	192.168.230.5	IMAP	85	Request: 5 list "" ...
47	6.624534268	192.168.230.5	192.168.230.4	IMAP	144	Response: * LIST (\...
48	6.632535728	192.168.230.4	192.168.230.5	IMAP	85	Request: 6 list "" ...
49	6.633615702	192.168.230.5	192.168.230.4	IMAP	109	Response: 6 OK List...
50	6.644297612	192.168.230.4	192.168.230.5	IMAP	84	Request: 7 create "
51	6.649439629	192.168.230.5	192.168.230.4	IMAP	111	Response: 7 OK Crea...
52	6.658288695	192.168.230.4	192.168.230.5	IMAP	87	Request: 8 subscri...



No.	Time	Source	Destination	Protocol	Length	Info
53	6.666636023	192.168.230.5	192.168.230.4	IMAP	114	Response: 8 OK Subs...
54	6.667973779	192.168.230.4	192.168.230.5	IMAP	85	Request: 9 list "" ...
55	6.669636950	192.168.230.5	192.168.230.4	IMAP	151	Response: * LIST (\...
56	6.707132578	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
57	6.765120190	192.168.230.4	192.168.230.5	IMAP	85	Request: 10 select ...
58	6.766374377	192.168.230.5	192.168.230.4	IMAP	398	Response: * FLAGS (...
59	6.766750619	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
60	6.810648067	192.168.230.4	192.168.230.5	IMAP	75	Request: 11 IDLE
61	6.811608731	192.168.230.5	192.168.230.4	IMAP	76	Response: + idling
62	6.850643445	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
69	6.843732306	192.168.230.5	192.168.230.4	IMAP	83	Response: * OK Stil...
70	6.843906574	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
71	6.846800616	192.168.230.4	192.168.230.5	IMAP	72	Request: DONE
72	6.848856050	192.168.230.5	192.168.230.4	IMAP	89	Response: 11 OK Idl...
73	6.849570887	192.168.230.4	192.168.230.5	IMAP	75	Request: 12 noop
74	6.851942049	192.168.230.5	192.168.230.4	IMAP	89	Response: 12 OK NOO...
75	6.868633573	192.168.230.4	192.168.230.5	IMAP	75	Request: 13 IDLE
76	6.870386135	192.168.230.5	192.168.230.4	IMAP	76	Response: + idling
77	6.910653964	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
88	186.902041843	192.168.230.5	192.168.230.4	IMAP	83	Response: * OK Stil...
89	186.902201267	192.168.230.4	192.168.230.5	TCP	66	38580 → 143 [ACK] S...
90	186.904904440	192.168.230.4	192.168.230.5	IMAP	72	Request: DONE
91	186.906504460	192.168.230.5	192.168.230.4	IMAP	89	Response: 13 OK Idl...
92	186.907229137	192.168.230.4	192.168.230.5	IMAP	75	Request: 14 noop

imágenes 57 y 58: intercambios entre el servidor de correo IMAP y el cliente

Aquí podemos ver los el intercambio de mensajes del Handshake Protocol entre el servidor y el cliente haciendo una consulta https a www.nombreorganizacion.org, centrándonos especialmente en los mensajes de autenticación del cliente (mensaje Certificate Request por parte del servidor al cliente y el Certificate Verify del cliente al servidor)

No.	Time	Source	Destination	Protocol	Length	Info
31	11.933496308	192.168.230.4	192.168.230.5	TCP	66	42760 → 443 [ACK] S...
32	11.933903841	192.168.230.4	192.168.230.5	TLSv1.2	279	Client Hello
33	11.934482070	192.168.230.5	192.168.230.4	TCP	66	443 → 42760 [ACK] S...
34	11.937408730	192.168.230.5	192.168.230.4	TLSv1.2	2565	Server Hello, Certi...
35	11.937424697	192.168.230.4	192.168.230.5	TCP	66	42760 → 443 [ACK] S...
40	21.954007921	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
41	21.954977950	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
46	31.969704727	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
47	31.970863154	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
48	41.986481551	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
49	41.987474855	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
50	52.001608148	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
51	52.002651595	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
54	62.017834747	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
55	62.018921981	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
58	72.034245772	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
59	72.035229072	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
60	74.774831227	192.168.230.4	192.168.230.5	TLSv1.2	1490	Certificate, Client...
61	74.777310370	192.168.230.5	192.168.230.4	TLSv1.2	1380	New Session Ticket,...
62	74.777329073	192.168.230.4	192.168.230.5	TCP	66	42760 → 443 [ACK] S...
63	74.796327197	192.168.230.4	192.168.230.5	TLSv1.2	97	Encrypted Alert
64	74.797442460	192.168.230.5	192.168.230.4	TLSv1.2	97	Encrypted Alert
65	74.797448586	192.168.230.5	192.168.230.4	TCP	66	443 → 42760 [FIN, A...
66	74.798426794	192.168.230.4	192.168.230.5	TCP	66	42760 → 443 [RST, A...

Length: 167
 ▶ Handshake Protocol: Certificate Request
 ▶ Handshake Protocol: Server Hello Done

No.	Time	Source	Destination	Protocol	Length	Info
31	11.933496308	192.168.230.4	192.168.230.5	TCP	66	42760 → 443 [ACK] S...
32	11.933903841	192.168.230.4	192.168.230.5	TLSv1.2	279	Client Hello
33	11.934482070	192.168.230.5	192.168.230.4	TCP	66	443 → 42760 [ACK] S...
34	11.937408730	192.168.230.5	192.168.230.4	TLSv1.2	2565	Server Hello, Certi...
35	11.937424697	192.168.230.4	192.168.230.5	TCP	66	42760 → 443 [ACK] S...
40	21.954007921	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
41	21.954977950	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
46	31.969704727	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
47	31.970863154	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
48	41.986481551	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
49	41.987474855	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
50	52.001608148	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
51	52.002651595	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
54	62.017834747	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
55	62.018921981	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
58	72.034245772	192.168.230.4	192.168.230.5	TCP	66	[TCP Keep-Alive] 42...
59	72.035229072	192.168.230.5	192.168.230.4	TCP	66	[TCP Keep-Alive ACK]...
60	74.774831227	192.168.230.4	192.168.230.5	TLSv1.2	1490	Certificate, Client...
61	74.777310370	192.168.230.5	192.168.230.4	TLSv1.2	1380	New Session Ticket,...
62	74.777329073	192.168.230.4	192.168.230.5	TCP	66	42760 → 443 [ACK] S...
63	74.796327197	192.168.230.4	192.168.230.5	TLSv1.2	97	Encrypted Alert
64	74.797442460	192.168.230.5	192.168.230.4	TLSv1.2	97	Encrypted Alert
65	74.797448586	192.168.230.5	192.168.230.4	TCP	66	443 → 42760 [FIN, A...
66	74.798426794	192.168.230.4	192.168.230.5	TCP	66	42760 → 443 [RST, A...

Length: 1368
 ▶ Handshake Protocol: Certificate
 ▶ Handshake Protocol: Client Key Exchange
 ▶ Handshake Protocol: Certificate Verify

0000 08 00 27 da a9 89 08 00 27 88 18 17 08 00 45 00 ..'.....'.....E.
 0010 05 c4 30 71 40 00 40 06 b7 67 c0 a8 e6 04 c0 a8 ..0q@.0. .g.....

enp0s8: <live capture in progress> Packets: 105 · Displayed: 34 (32.4%) Profile: Default

imágenes 59 y 60: intercambio de mensajes del Handshake Protocol entre el servidor y el cliente

Traza del funcionamiento de FTP (además se puede apreciar cómo aparece el Authentication Header, que demuestra que el paquete FTP está encapsulado y protegido por IPSec):

ip.addr == 192.168.230.4 and ip.addr == 192.168.230.5						
No.	Time	Source	Destination	Protocol	Length	Info
50	3.145798990	192.168.230.5	192.168.230.4	FTP	113	Response: 230 ...
51	3.146060374	192.168.230.4	192.168.230.5	TCP	90	39466 → 21 [AC...
54	3.147445549	192.168.230.4	192.168.230.5	FTP	96	Request: SYST
56	3.148282545	192.168.230.5	192.168.230.4	FTP	109	Response: 215 ...
62	3.186571362	192.168.230.4	192.168.230.5	TCP	90	39466 → 21 [AC...
123	15.950500477	192.168.230.4	192.168.230.5	FTP	98	Request: TYPE I
125	15.951377643	192.168.230.5	192.168.230.4	FTP	121	Response: 200 ...
127	15.951941918	192.168.230.4	192.168.230.5	TCP	90	39466 → 21 [AC...
128	15.952331669	192.168.230.4	192.168.230.5	FTP	118	Request: PORT ...
129	15.953299318	192.168.230.5	192.168.230.4	FTP	141	Response: 200 ...
132	15.954724912	192.168.230.4	192.168.230.5	FTP	112	Request: STOR ...
135	15.955775370	192.168.230.5	192.168.230.4	TCP	98	20 → 50681 [SY...
136	15.955799784	192.168.230.4	192.168.230.5	TCP	98	50681 → 20 [SY...
137	15.956233700	192.168.230.5	192.168.230.4	TCP	90	20 → 50681 [AC...
138	15.956919063	192.168.230.5	192.168.230.4	FTP	112	Response: 150 ...
139	15.957223967	192.168.230.4	192.168.230.5	FTP-DA...	117	FTP Data: 27 b...
140	15.957566702	192.168.230.4	192.168.230.5	TCP	90	50681 → 20 [FT...
▶ Internet Protocol Version 4, Src: 192.168.230.4, Dst: 192.168.230.5 ▶ Authentication Header ▶ Transmission Control Protocol, Src Port: 50681, Dst Port: 20, Seq: 1, Ack: 1, Len: 27 FTP Data (27 bytes data) [Setup frame: 128] [Setup method: PORT] [Command: STOR prueba2_ftp.txt] Command frame: 132 [Current working directory:] ▼ Line-based text data (1 lines) Prueba de subida 2 por FTP\n						

imagen 61: intercambio FTP y muestra del mensaje FTP Data con el contenido del archivo